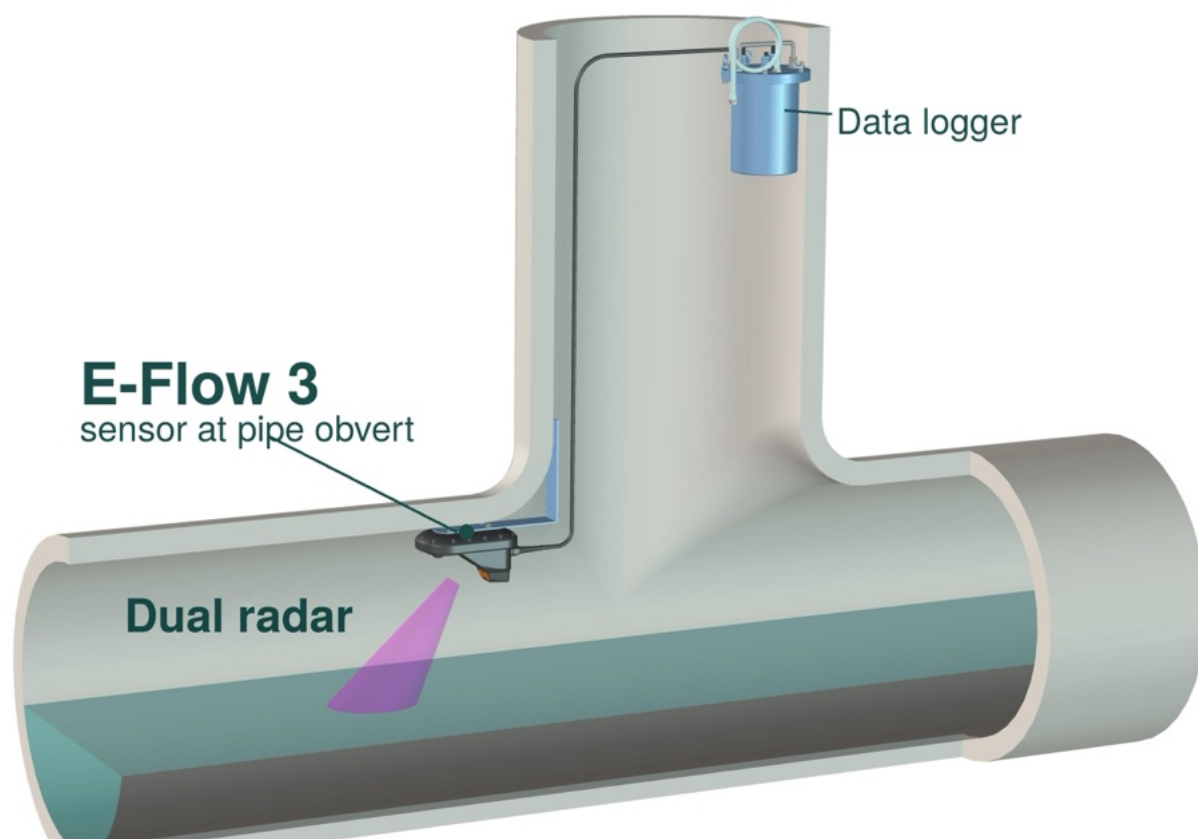


DUAL MODE SEWER FLOWMETER

E-Flow 3

The ultimate maintenance free sewer flowmeter.
Dual radar for everyday flow, dual ultrasonic
and sealed pressure for surcharge, in one
sensor.



Nothing in the flow

Contactless radar measures velocity and depth from the pipe obvert in normal flow, so grease, rag and silt never reach the sensor.

No gap when it surcharges

Dual 1 MHz ultrasonic velocity and a sealed 316 stainless pressure transducer take over seamlessly when the pipe runs full.

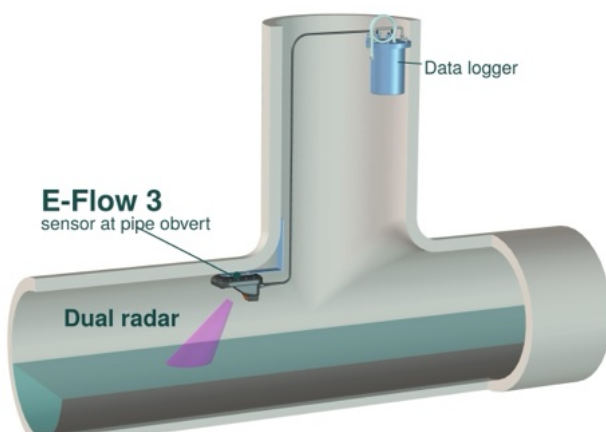
Built for the sewer

Fully potted, corrosion resistant housing, installed and commissioned from the surface without confined space entry.

FOUR SENSORS, TWO PAIRS, ONE RECORD

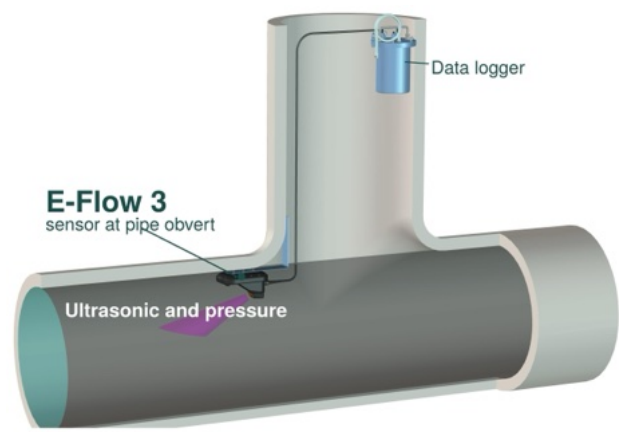
Two ways to measure. No reason to enter the sewer.

Every sewer flowmeter faces the same problem: the conditions that matter most, wet weather and surcharge, are the conditions that foul a wetted sensor. E-Flow 3 answers it with two sensor pairs in one body, each used where it works best, with seamless changeover between them.



Normal flow: dual radar

A 24 GHz Doppler radar reads surface velocity and a 120 GHz radar reads depth, both from above the flow. For the great majority of the year the sensor has no wetted measuring element at all, so there is nothing for rag to snag on, nothing for grease to coat and nothing to clean.



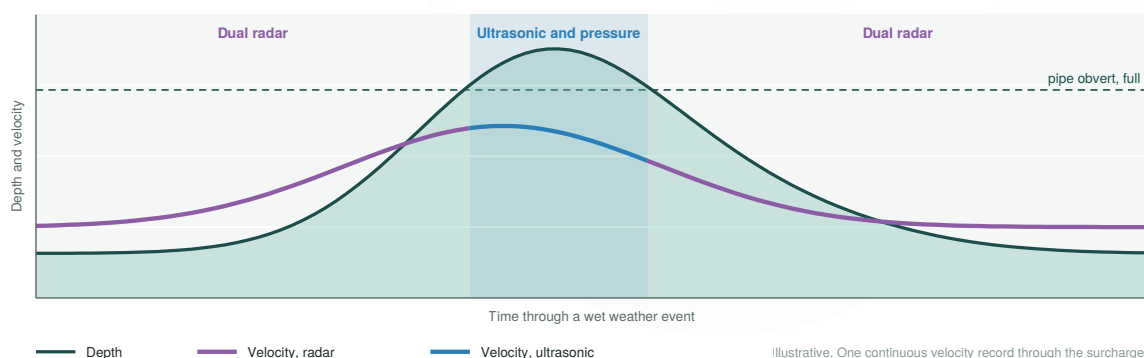
Surcharge: ultrasonic and pressure

When the sewer surcharges, radar loses its surface. E-Flow 3 changes seamlessly to its dual 1 MHz ultrasonic Doppler velocity sensor and a sealed 316 stainless steel high accuracy pressure transducer, and keeps measuring velocity and depth through the full water column, with no blind zone between modes.

The record you actually need in wet weather

Peak wet weather flow, overflow volumes and surcharge duration are the numbers that drive inflow and infiltration programs, capacity assessments and regulatory reporting. They are also exactly when a conventional wetted sensor is most likely to be fouled or reading badly.

E-Flow 3 keeps measuring through the whole event: radar as the level rises, ultrasonic and pressure through the surcharge, radar again as it recedes. The bidirectional model also flags reverse flow and backing up from downstream restrictions.

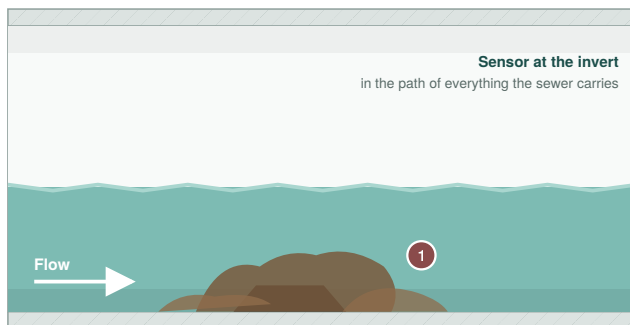


MAINTENANCE FREE IN PRACTICE

Where the sensor sits decides how often you visit it.

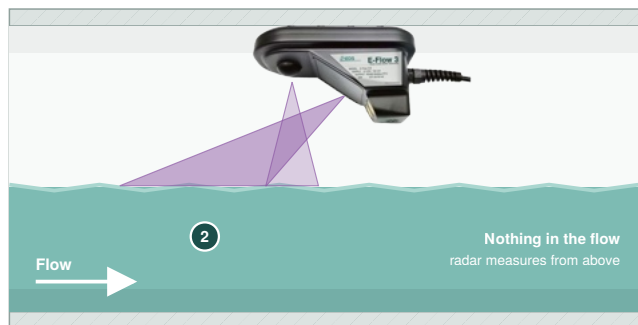
Most sewer flowmeters live at the invert, in the path of everything the sewer carries. E-Flow 3 lives at the obvert, above the flow, and only touches the water when the pipe is full.

Conventional invert mounted sensor



- 1 Rag, grease and silt accumulate on the sensor and the data drifts.

E-Flow 3 at the obvert



- 2 Solids pass beneath. No wetted velocity element in normal service.

What fouling costs

A wetted velocity sensor at the invert collects rag and grease within weeks in a busy sewer. The velocity reading drifts, the depth reading is disturbed, and the record has to be corrected or discarded. Every clean is a confined space entry or a retrieval, and a gap in the data.

What obvert mounting changes

With the radar pair measuring from above the flow there is no wetted velocity element to foul in normal service. The ultrasonic pair and pressure transducer are only in contact during surcharge, and the potted body sheds material once the level drops. Site visits become verification, not cleaning.

HOW DOPPLER VELOCITY WORKS

1 Transmit



The sensor sends a signal of known frequency toward the moving flow.

2 Reflect



Particles and the surface moving toward the sensor return the signal compressed, a higher frequency.

3 Measure



The shift between sent and returned frequency is proportional to velocity. $\Delta f \propto v$

Both velocity technologies in E-Flow 3 use the same Doppler principle. Radar applies it to the moving surface; ultrasonic applies it to solids and bubbles carried within the flow. The sensor then converts the measured velocity to mean channel velocity and, with depth and pipe profile, to flow rate.

No cleaning visits

in normal service. The velocity and depth radars have no wetted element to foul.

No confined space entry

to install, commission or verify. Everything is done from the maintenance hole opening.

One continuous record

through surcharge, with the active sensor pair reported alongside the data.

ENGINEERED FOR THE ENVIRONMENT

Designed for harsh, demanding sewer environments

Hydrogen sulphide, grease, rag, silt, temperature swings and the occasional full surcharge. E-Flow 3 is engineered to sit in that environment for years and keep reporting.



Contactless in normal flow

Dual radar measures velocity and depth from above the flow. No wetted element for most of the year.



Seamless through surcharge

Dual 1 MHz ultrasonic velocity and a sealed 316 stainless pressure transducer take over automatically.



Fully potted, sealed for life

Alloy and engineering plastic composite housing, fully potted electronics, unaffected by temperature cycling.



Obvert mounting sheds debris

Mounted at the top of the pipe on an EDS bracket, out of the flow path, so material cannot build up on the sensor face.



No confined space entry

Installed and retrieved from the maintenance hole opening; commissioned and verified from the surface with EMS-Flow.



Diagnostics you can trust

Each of the four sensors reports its own value and status, so you can see which pair is active and why, from the office.

Where E-Flow 3 fits

Inflow and infiltration

Reliable wet weather velocity and depth through surcharge, the events that define an I&I program.

Network monitoring

Long term flow at key points in reticulation and trunk sewers with minimal site visits.

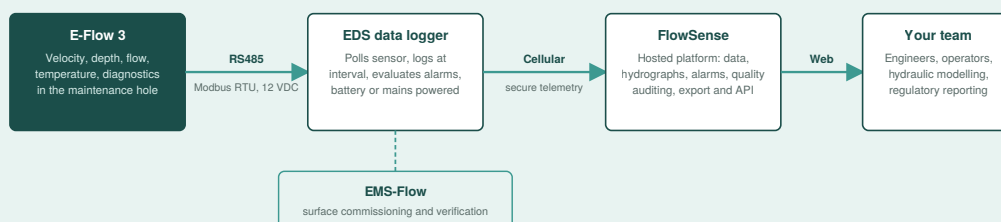
Overflow and surcharge

Capture the peak, the duration and, with the bidirectional model, any reverse flow.

Pump station and catchment

Inflow to wet wells and catchment boundaries, feeding capacity and growth planning.

Part of the EDS system



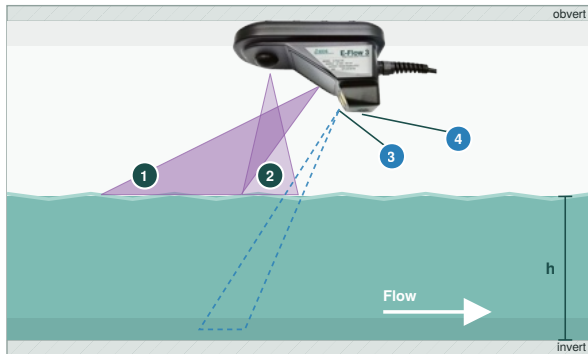
Connect E-Flow 3 to an EDS logger and the FlowSense telemetry platform for hosted data, surcharge and reverse flow alarms, data quality auditing and export to your hydraulic modelling tools.

EDS installs, commissions and supports from Brisbane, with field crews and rental fleets for temporary studies across Australia. Australian owned and operating in the water and wastewater industry since 1991.

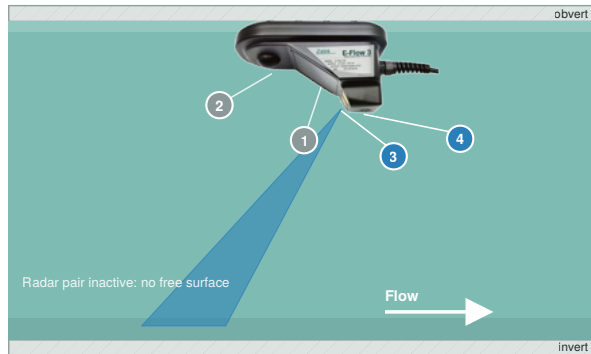
PRINCIPLE OF OPERATION

Specifications at a glance

Normal flow, partially full



Surcharged, pipe full



- 1 Velocity radar 2 Depth radar 3 Dual ultrasonic velocity 4 Pressure transducer

VELOCITY

Method	Contactless 24 GHz radar in normal flow; dual 1 MHz contact ultrasonic Doppler in surcharge; seamless changeover
Range	0.05 to 20 m/s, recommended above 0.2 m/s
Accuracy	Ultrasonic 1 % ± 0.005 m/s; radar 2 % ± 0.01 m/s
Direction	Unidirectional or bidirectional model

DEPTH

Method	120 GHz FMCW radar in normal flow; sealed 316 stainless steel high accuracy pressure transducer in surcharge
Range	0 to 10 m
Accuracy	0.2 % FS ± 0.001 m

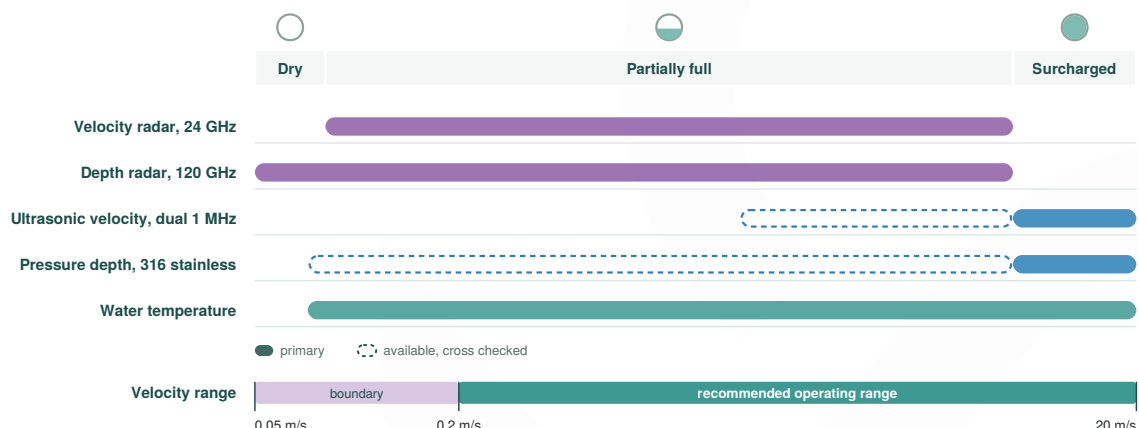
ELECTRICAL

Power	8 to 15 VDC, 150 mA at 12 VDC
Output	RS485 Modbus RTU, configurable damping, 10 s response
Temperature	0 to 60 °C water temperature, 0.5 °C accuracy

MECHANICAL

Housing	Alloy and engineering plastic composite, fully potted
Mounting	Pipe obvert on EDS bracket, from the maintenance hole opening
Models	E-Flow 3 U (unidirectional), E-Flow 3 B (bidirectional)

Full specifications are in the E-Flow 3 datasheet, EF3 DS 001.



Which sensor is active as the pipe fills. Solid: primary. Dashed: available and cross checked.

E-FLOW 3 SENSOR

Four sensors, one potted body, built for the sewer.



E-Flow 3 with integral cable and obvert mounting face

Alloy and engineering plastic composite housing, fully potted electronics

TALK TO EDS ABOUT E-FLOW 3

1300 721 683

eds@e-d-s.com.au

www.e-d-s.com.au

Environmental Data Services

13/20 to 22 Ellerslie Road, Meadowbrook QLD 4131

Tell us the pipe size, the depth and what you need from the data. We will size the bracket and cable, recommend the logger and telemetry, and can install and commission for you. For temporary studies, E-Flow 3 is available with EDS rental loggers and FlowSense.

Australian owned and operating in the water and wastewater industry since 1991. Installation, commissioning and support from Brisbane, with field crews across Australia.